Example 3.9

Find the roots of the following polynomial equations.

(i) $5x - 3 = 0$ (ii) $-7 - 4x = 0$

Solution

(i) $5x - 3 = 0$

(or) $5x = 3$

Then, $x = \frac{3}{5}$

(ii) $-7 - 4x = 0$

(or) $4x = -7$

Then, $x = \frac{7}{-4} = -\frac{7}{4}$

Note

(i) A zero of a polynomial can be any real number not necessarily zero.

(ii) A non zero constant polynomial has no zero.

(iii) By convention, every real number is zero of the zero polynomial

Example 3.10

Check whether -3 and 3 are zeros of the polynomial $x^2 - 9$

Solution

Let $f(x) = x^2 - 9$

Then, $f(-3) = (-3)^2 - 9 = 9 - 9 = 0$

$f(+3) = 3^2 - 9 = 9 - 9 = 0$

$\therefore$ -3 and 3 are zeros of the polynomial $x^2 - 9$

Exercise 3.2

1. Find the value of the polynomial $f(y) = 6y - 3y^2 + 3$ at

(i) $y = 1$ (ii) $y = -1$ (iii) $y = 0$

2. If $p(x) = x^2 - 2\sqrt{2}x + 1$, find $p(2\sqrt{2})$.

3. Find the zeros of the polynomial in each of the following :

(i) $p(x) = x - 3$ (ii) $p(x) = 2x + 5$ (iii) $q(y) = 2y - 3$

(iv) $f(z) = 8z$ (v) $p(x) = ax$ when $a \neq 0$ (vi) $h(x) = ax + b,\ a \neq 0,\ a, b \in R$

4. Find the roots of the polynomial equations .

(i) $5x - 6 = 0$ (ii) $x + 3 = 0$

(iii) $10x + 9 = 0$ (iv) $9x - 4 = 0$

5. Verify whether the following are zeros of the polynomial indicated against them, or not.

(i) $p(x) = 2x - 1,\ x = \frac{1}{2}$ (ii) $p(x) = x^3 - 1,\ x = 1$

(iii) $p(x) = ax + b, x = \frac{-b}{a}$ (iv) $p(x) = (x + 3)(x - 4),\ x = 4, x = -3$

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